

Employment

Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Shipped 10+ production TypeScript/Node.js microservices end-to-end: REST APIs, real-time event pipelines, multi-model AI agent tooling (Claude/GPT-4o/Gemini), and Auth0 RBAC; 923 commits via GitHub Actions CI/CD across 7 deployments. - Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); maintain CI/CD and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deploy genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. - Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Inventory Reservation Service	Java 21 + Spring Boot	2026
- Built transactional inventory reservation API in Java 21 + Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations; idempotent endpoints with RFC 9457 error responses and automated expiry via scheduled cleanup. - Verified concurrency correctness with Testcontainers integration tests; Flyway migrations, Spring Security RBAC, Docker, GitHub Actions CI.		
VectorLift	Semantic Search & Indexing Pipeline	2026
- Built production semantic search on MS MARCO: Elasticsearch BM25 baseline, sentence-transformers bi-encoder fine-tuned with in-batch negatives, MiniLM cross-encoder re-ranker; benchmarked 6 pipeline configurations with NDCG@10, MRR@10, MAP, and bootstrap significance tests. - Designed Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; served search via FastAPI with per-stage latency tracking and Streamlit experiment dashboard.		
PulseRank	Event-Driven Leaderboard & Feed	2026
- Built event-driven leaderboard in TypeScript/NestJS: MySQL with idempotency-key deduplication and outbox pattern, MongoDB TTL projections, Redis sorted sets for sub-millisecond reads; SSE live push. - Validated multi-store consistency with Testcontainers; benchmarked p99 latency with k6; GitHub Actions CI gates.		

Languages and Technologies

- Languages:** Java, TypeScript, Python, JavaScript, SQL, Bash
- Backend & Data:** Spring Boot, Node.js, NestJS, FastAPI, PostgreSQL, MySQL, MongoDB, Redis, Kafka
- Frontend:** React, React Native, Next.js, Tailwind CSS, SSE, WebSockets
- Search & NoSQL:** Elasticsearch, Qdrant, Cassandra (familiar), Couchbase (familiar)
- Cloud & DevOps:** GCP, AWS, Docker, Kubernetes, Terraform, GitHub Actions, OpenTelemetry, Grafana

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs within 48-hour sprints, each with a working demo.